

Unmanned Aircraft Systems

UAS

Loading and Performance

FAA Remote Pilot Part 107 Exam Prep

FAA Remote Pilot Test Prep

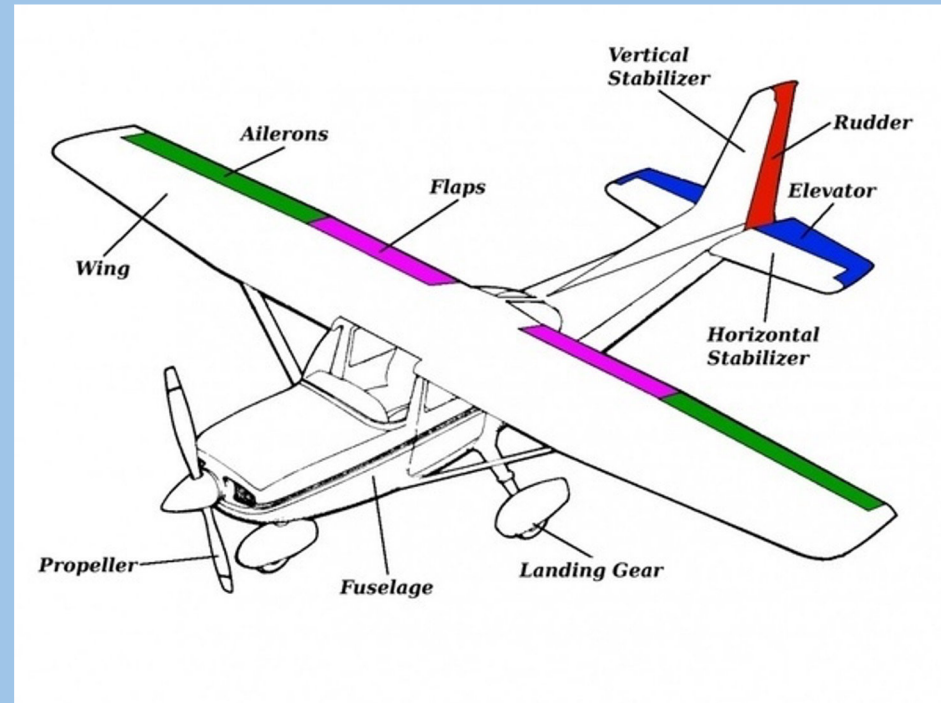
Topics Covered in this lesson

- Introduction
- Determining speed and altitude
- Loading
- Computing weight and balance
- Load factor
- Stalls
- Performance



Loading and Performance Introduction

- ◆ Prior to each flight, the remote PIC must ensure that any object attached to or carried by the small unmanned aircraft is secure and does not adversely affect the flight characteristics or controllability of the aircraft.



Loading and Performance

Introduction

Primary Control Surface	Airplane Movement	Axes of rotation	Type of stability
Aileron	Roll	Longitudinal	Lateral
Elevator/Stabilator	Pitch	Lateral	Longitudinal
Rudder	Yaw	Vertical	Directional

Loading and Performance

Determining Speed and Altitude

- ◆ **Dead reckoning** – navigation solely by means of computations based on time, airspeed, distance, and direction. The products derived from these variables, when adjusted by wind speed and velocity, are heading and ground speed.
- ◆ **Pilotage** – navigation by reference to landmarks or checkpoints.
- ◆ **The Wind Triangle** – navigation using triangulation. The true heading and ground speed can be found by drawing a wind triangle of vectors. One side of the triangle is the wind direction and velocity, another side is the true heading and true airspeed, the last side is the track, or true course, and the ground speed. Each side of a wind triangle is the vector sum of the other two sides.

Loading and Performance

Determining Speed and Altitude

- ◆ Altitude determination on a sUAS: [sUAS must remain below 400 feet AGL (Above Ground Level)]
 - ▣ Install a calibrated altimeter that reports MSL (mean sea level) altitude back to control station. AGL (above ground level) altitude must be converted by adding or subtracting field elevation from MSL altitude.
 - ▣ GPS reported altitude from sUAS.
 - ▣ Have a visual observer pace off 400' to establish a rough visual reference of 400 feet.
 - ▣ Use the height of known terrain or obstructions.
 - ▣ Use a measurement of distance if control station is so equipped.

Loading and Performance

Loading

- ◆ **Empty weight** - includes the airframe, power source, all fixed equipment, and unusable fuel.
- ◆ **Useful load** – power source (battery or fuel) and payload equipment.
- ◆ **Launch weight** – empty weight plus useful load.
- ◆ **Landing weight** – launch weight minus any fuel used or jettisoned load.

Loading and Performance

Computing weight and balance

- ♦ **Arm** - horizontal distance measured in inches from the datum line to a point on the aircraft.
- ♦ **Datum line** - a reference point along the longitudinal axis indicated by the manufacturer.
- ♦ **Moment** – product of the weight of an object multiplied by its arm and is expressed in pound-inches (lbs.-in).
- ♦ **C.G. (center of gravity)** – The total moment divide by the total weight equals the CG.
C.G. represents the number of inches from the datum that the airplanes balances.
- ♦ While longitudinal stability is typically the focus of weight and balance , lateral CG is also a concern. (eg. fuels used in the wings must be used evenly or the aircraft can become laterally unstable).

Loading and Performance

Computing weight and balance

- ♦ Loading an aircraft outside of its CG limits can make the aircraft difficult or impossible to control.
- ♦ It is the responsibility of the remote PIC to use the most current weight and balance data when planning a flight and operating the sUAS.

Example of a weight and balance calculation

	Weight	x	Arm	=	Moment/1000
Aircraft empty weight:	1639lbs		39.29		64.4 in/lbs.
Pilot & Front Passenger:	340 lbs.		37"		12.58 in/lbs.
Fuel: 53 gallons (max)	318 lbs.		48"		15.26 in/lbs.
Rear passenger	80 lbs.		72"		5.76 in/lbs.
Baggage:	30 lbs.		76"		2.28
<u>Total:</u>	<u>2,407 lbs.</u>				<u>100.28</u>

Center of gravity = total moment/total weight = $[100.28/2407 \times 1000] = \underline{41.66"} \text{ from Datum}$

Loading and Performance

Load Factor

- ♦ **Load** - in aerodynamics, load is the force or imposed stress that must be supported by a sUAS structure in flight.
- ♦ **Load Factor** – The loads imposed on the wings or rotors in flight.
- ♦ **G force** – the force an object or person experiences due to acceleration.
- ♦ An aircraft in straight and level unaccelerated flight experiences 1 G. Gravity applies and acceleration force of 1 G.
- ♦ G force is increased or decreased in turns, climbs, dives, accelerations, and decelerations.

Loading and Performance

Load Factor

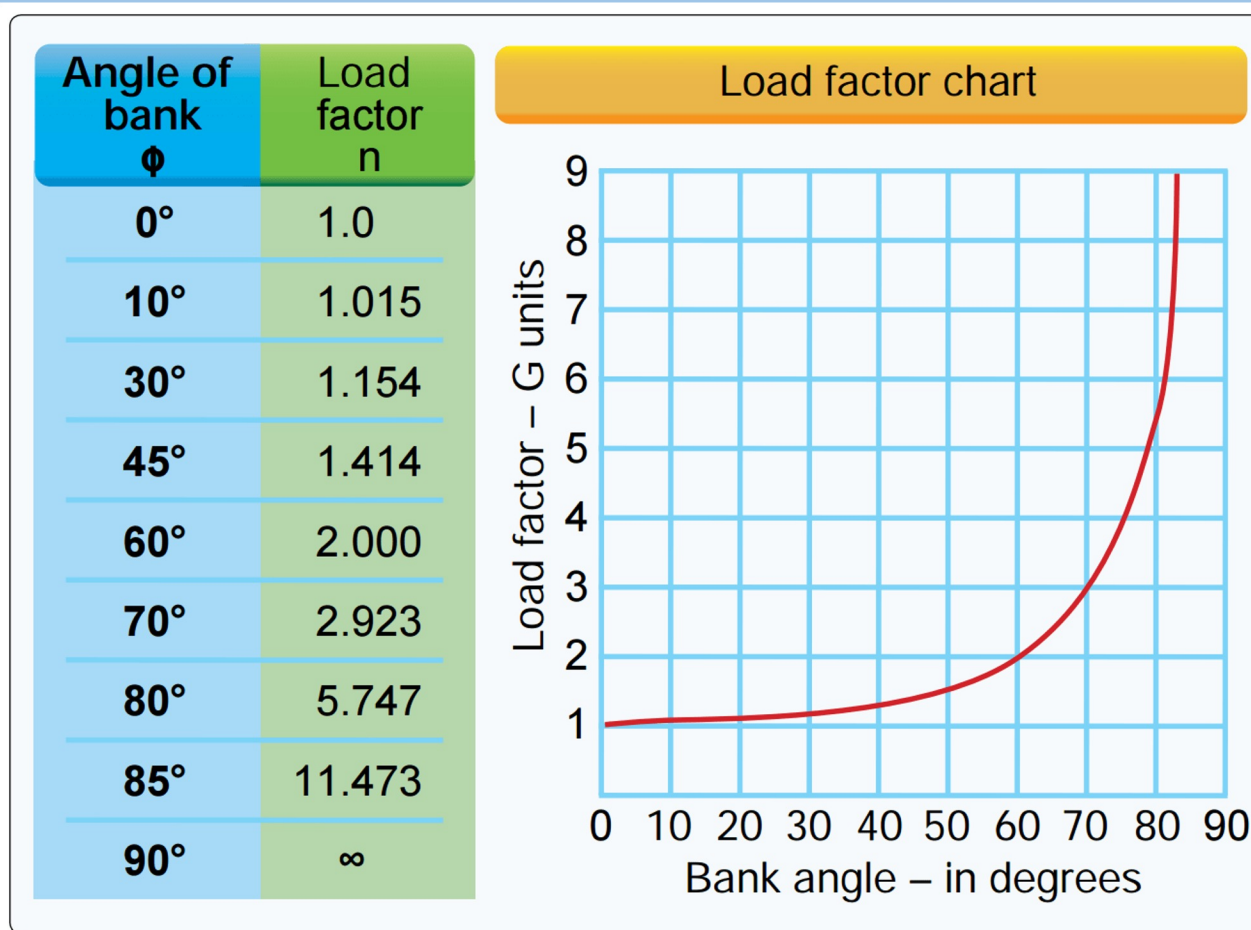
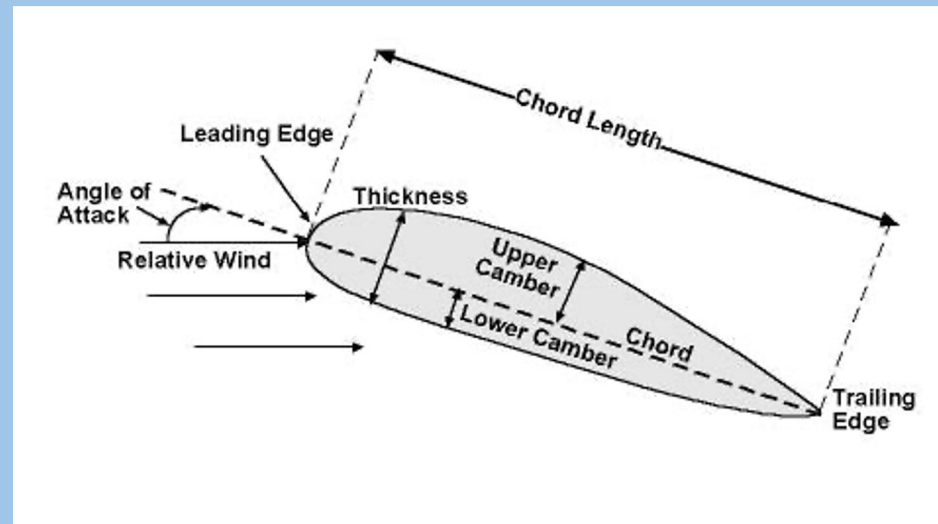


FIGURE 2.—Load Factor Chart.

Loading and Performance

Stalls

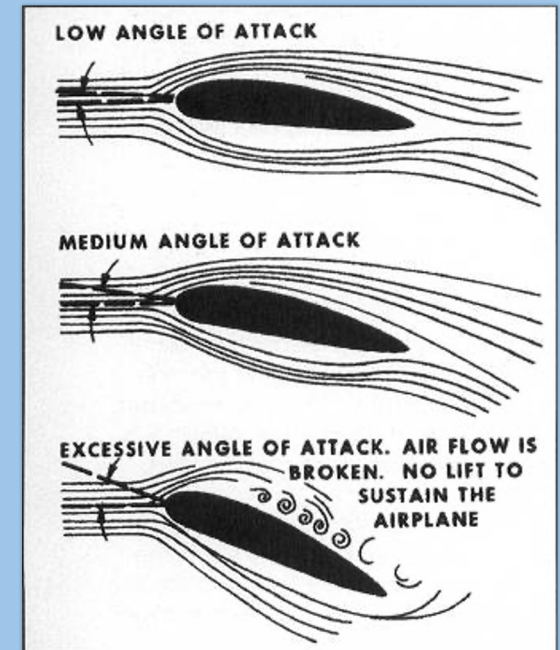
- ♦ **Airfoil** – a structure or body that produces a useful reaction to air movement.
- ♦ **Chord Line** – an imaginary straight line from the leading edge to the trailing edge of an airfoil.
- ♦ **Relative wind** - the wind “felt” or experienced by an airfoil.
- ♦ **Angle of attack** – angle between the chord line of the airfoil and the relative wind.



Loading and Performance

Stalls

- Stalls occur when the critical angle of attack is exceeded (18-20 degrees for most airfoils). During a stall, the smooth airflow over the airfoil becomes separated from the airfoil.
- Turbulence caused by the separation of airflow becomes so great the airfoil stops producing enough lift to maintain flight.
- A wing will always stall at the critical angle of attack regardless of altitude, airspeed, or weight.
- Increases in load factor will increase an aircraft's stall speed.



Loading and Performance

Performance

- ◆ The remote PIC is responsible for ensuring that every flight can be accomplished safely, does not pose an undue hazard, and does not increase the likelihood of a loss of positive control.
- ◆ The effect of runway slope on launch distance is due to the component of weight along the inclined path of the aircraft. An upslope would contribute a retarding force component, while a downslope would contribute an accelerating force component.
- ◆ Weight, altitude, temperature and unfavorable wind are the most critical factors in launch performance of a sUAS.
- ◆ The maximum endurance condition is obtained at the point of minimum power required (least fuel or battery power required to maintain flight).

References

- ◆ Marshall, D.; Barnhart, R., Shappee, E., Most, M. (2016). *Introduction to Unmanned Aircraft Systems*. New York, NY: CRC Press.